

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2026 SECONDARY 4 EXPRESS

ADDITIONAL MATHEMATICS

4049/02

Paper 2

24 August 2026

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total marks for this paper is 90.

Total Marks

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formula for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}bc \sin A$$

1 (a) Show that $\cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$. [4]

(b) Hence find an expression for $\sin^2 75^\circ$ in the form $\frac{a + b\sqrt{3}}{4}$ where a and b are integers. [3]

- 2 Show that there is no real value of k for which the curve $y = kx^2 + (k - 1)x + 4$ lies completely below the line $y = 2x + 3$. [5]

- 3 Solve the equation $2 \tan^2 x + \sec x - 4 = 0$ for $0 \leq x \leq 2\pi$. [6]

- 4 A culture of bacteria is grown under laboratory conditions.

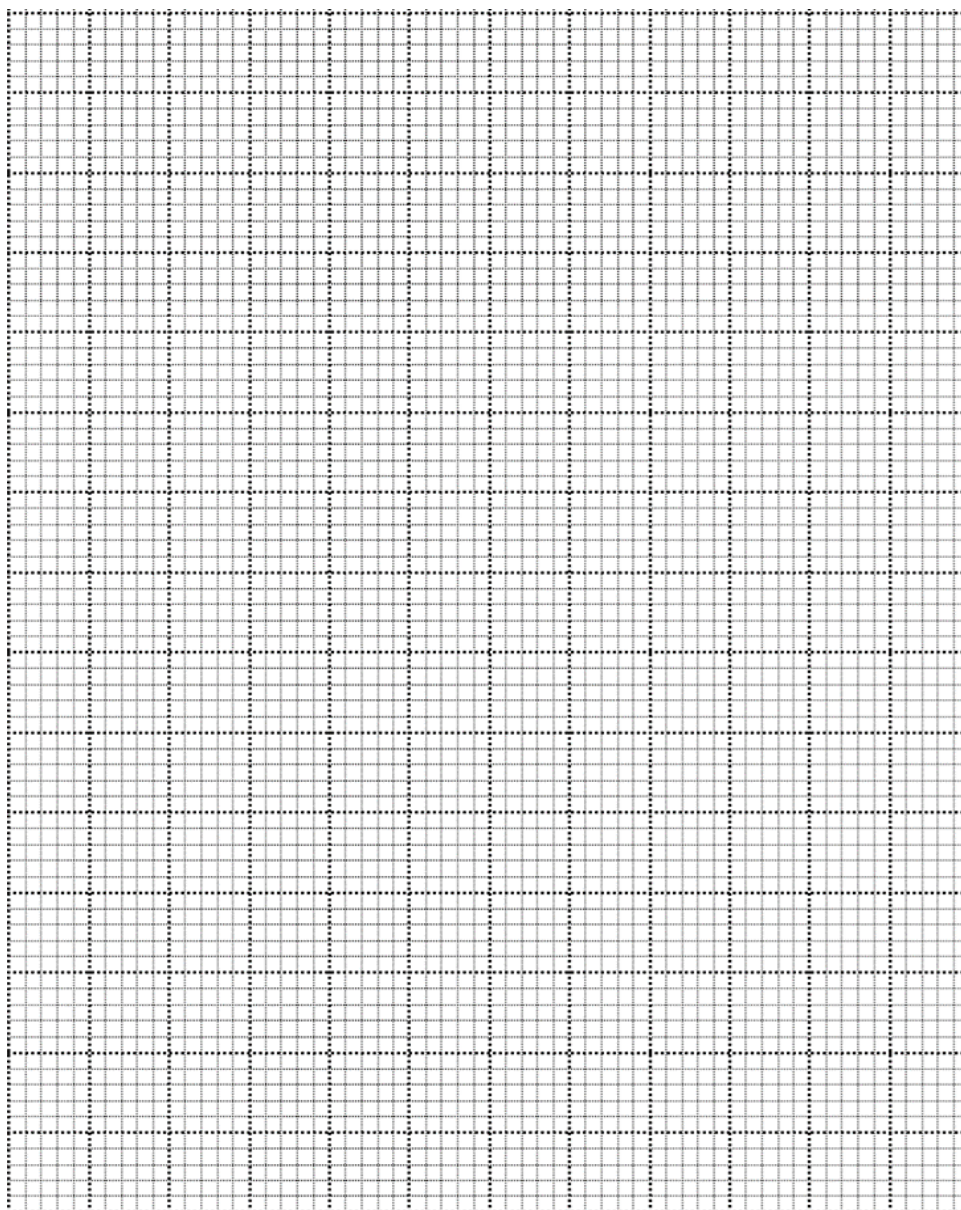
The number of bacteria, N , after t hours is believed to be related by the equation $N = ab^t$, where a and b are constants.

The table below shows the number of bacteria, N , after different values of t .

t	1	2	3	4	5
N	1200	3600	10 800	32 400	97 200

- (a) On the grid below, plot $\ln N$ against t and draw a straight line graph.

[3]



Use your graph to estimate

(b) the value of a ,

[2]

(c) the value of b .

[2]

5 A solar panel is installed on a rooftop.

The average daily energy output, E , in kWh, is modelled by $E = -0.01\theta^2 + 0.4\theta + 2$, where θ is the tilt angle at which the panel is tilted from the horizontal, and $0^\circ \leq \theta \leq 40^\circ$.

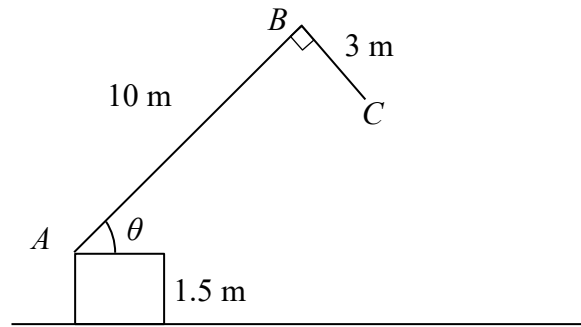
(a) Express E in the form $E = a + b(\theta - c)^2$, where a , b and c are constants to be found. [3]

(b) Hence state the maximum daily energy output by the solar panel and the tilt angle at which this occurs. [2]

(c) A technician claims that halving the optimal tilt angle found in part **(b)** will halve the average daily energy output.
Explain why the claim is incorrect. [1]

- 6 (a)** It is given that $f(x) = x^3 + px^2 + qx + 6$, where p and q are constants.
 $f(x)$ has a factor of $x - 2$, and leaves a remainder of 18 when divided by $x + 1$.
Find the values of p and q . [5]
- (b)** Given that $2x - 3$ is a factor of $P(x) = 2x^3 - 5x^2 + x + 3$, solve the equation $P(x) = 0$,
expressing all roots in exact form. [4]

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The diagram shows a truck mounted with an L-shape arm for lifting objects.

The L-shape arm is formed by AB and BC rigidly joined at B such that angle $ABC = 90^\circ$.

The lengths of AB and BC are 10 m and 3 m respectively.

The point A is fixed on the truck of 1.5 m tall and the L-shape arm rotates in a vertical plane with AB inclined at an angle of θ to the top of the truck which is parallel to the ground.

(a) Show that the height, h metres, of C above the ground is given by

$$h = 10 \sin \theta - 3 \cos \theta + 1.5.$$

[3]

(b) Express h in the form $p + R \sin(\theta - \alpha)$, where p is a constant, $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. [4]

(c) Find the value of θ if C is at a height of 5 m above the ground. [2]

- 8 The gradient of a curve at a point (x, y) is given by $\frac{dy}{dx} = \frac{a}{x} + bx - \frac{4}{x^2}$, where a and b are constants, and $x > 0$. The gradient has a stationary value at $x = 2$. The curve has a stationary point at $(1, 6)$. Find the equation of the curve. [9]

Extra working space

9 Points $A(-1, -2)$ and $B(7, 4)$ lie on a circle with centre C .

The tangent to the circle at B has equation $4x + 3y = 40$.

(a) Show that point A lies on the normal to the circle at B .

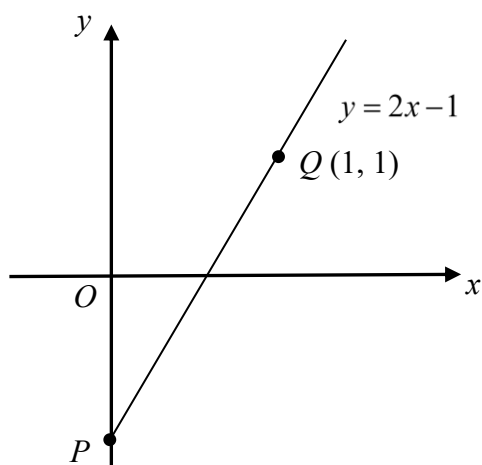
[3]

(b) Hence find the equation of the circle.

[3]

- (c) The tangent to the circle at another point D meets the tangent to the circle at B at a point E which lies vertically above the centre C .
Find the area of quadrilateral $BCDE$. [5]

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The diagram above shows a particle moving along the line $y = 2x - 1$.

It moves in such a way that the x -coordinate of the particle is increasing at a constant rate of 0.05 units per second from P to Q .

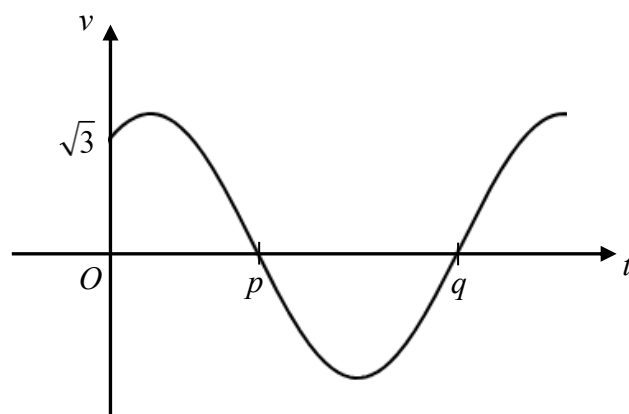
(a) Show that the distance of the particle from O , in terms of x , is given by $\sqrt{5x^2 - 4x + 1}$. [2]

(b) Find the rate at which the distance of the particle from O is increasing at the instant when the particle passes through Q . [3]

After the particle passes Q , the x -coordinate of the particle then increases at a rate of x^2 units per second. At a point R , the area of triangle OPR is increasing at a rate of at least 200 units² per second.

- (c) Find the possible set of values of the x -coordinate of R . [4]

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The diagram shows the velocity-time graph of a particle moving in a straight line.

The velocity, v m/s, of the particle, t seconds after passing the fixed point O , is given by

$v = k \cos\left(t - \frac{\pi}{6}\right)$. The initial velocity of the particle is $\sqrt{3}$ m/s.

(a) Show that $k = 2$. [2]

(b) Find the values of p and q in exact form. [2]

(c) Find the acceleration of the particle at $t = 3$. [2]

(d) Find the distance travelled by the particle from $t = 0$ to $t = q$. [5]

(e) What does the integral $\int_0^q k \cos\left(t - \frac{\pi}{6}\right) dt$ represent in this context? [1]

